L2a: Are Risk-Free Assets Really Risk-Free?

CHEME 5660 Cornell University

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Objectives for Today

Today we introduce U.S. marketable Treasury securities, develop their cash-flow and pricing conventions, and ask what risk-free does and does not mean.

Today's concepts

- Describe Treasury securities: distinguish regular bills, Cash Management Bills, notes, and bonds by their maturities and cash flows.
- Value Treasury cash flows: separate par from price and coupon rates from market yields, then use dated cash flows to calculate price.
- Interpret the risk-free benchmark: explain low default risk, changing market prices, and the flight to Treasuries during financial stress.

Companion notebook: L2a lecture notebook

Lectures and Examples

Lecture: CHEME-5660-L2a-Lecture-TreasurySecurities-Fall-2026.ipynb

The worked examples reconstruct reported auction prices from the Treasury's quotation conventions and actual auction data.

Example: CHEME-5660-L2a-TBills-Pricing-Fall-2026-WorkedExample.ipynb
Zero-coupon bill auctions, 2023–2026.

Example: CHEME-5660-L2a-NotesBonds-Pricing-Fall-2026-WorkedExample.ipynb
Coupon note and bond auctions, 2022–2026.

Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

Quick review before we start: the time value of money, abstract assets, and net present value.

Review: Yields and Accumulation Factors

Dated dollars have different units. A dollar today and a dollar at a future date cannot be added until a stated rule expresses them at a common valuation date.

Let P be an amount held today, y a quoted nominal annual yield, and n the compounding intervals per year. At period index j, the equivalent future amount is:

$$F_j = \underbrace{(1 + y/n)^j}_{\mathcal{D}_{j,0}(y;n)} P.$$

$\mathcal{D}_{j,0}(y;n)$ converts time-0 dollars into period-j dollars and its inverse converts them back. The continuously compounded equivalent is $g_y := n \log(1 + y/n)$, so $\mathcal{D}_{T,0}(y;n) = (1 + y/n)^{nT} = e^{g_y T}$ over T years.

Careful: g is a growth rate and $r = (\Delta t)\, g$ its log return. Convert a quoted y before comparing.

Review: Abstract Assets and Net Present Value

Abstract assets describe financial value as dated signed cash flows; each must be converted to a common valuation date before the values are added.

Let N be the final cash-flow index and $\bar{c}_j$ the signed net cash flow at index j. Under a constant nominal yield y, the time-0 net present value is:

$$\mathrm{NPV}_0(y) = \sum_{j=0}^{N} \mathcal{D}_{j,0}^{-1}(y;n)\,\bar{c}_j.$$

Every term in the sum is measured in time-0 dollars.

Today: set $\mathrm{NPV}_0 = 0$ and solve for the price. That is what pricing a Treasury security means.

The NPV Sign Is a Valuation Test

With every cash flow in today's dollars, NPV compares the present value received with the amount paid.

- **Positive NPV:** present value received $>$ amount paid; the asset clears the chosen discount-rate benchmark.
- **Negative NPV:** present value received $<$ amount paid; the asset does not clear that benchmark.
- **Zero NPV:** present value received $=$ amount paid; this is the pricing condition used for Treasury securities.

The sign is conditional on the discount model and assumed cash flows, not a promise about what will happen.

Treasury Bills

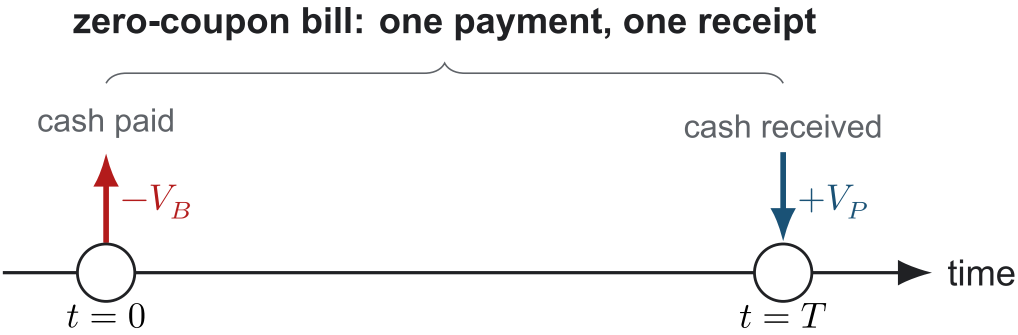
A U.S. Treasury bill is a short-term zero-coupon debt security: the investor receives no periodic coupon and is paid the par amount at maturity.

- **Regular Treasury bills** have standardized 4-, 6-, 8-, 13-, 17-, 26-, or 52-week terms on a recurring auction schedule, bought through TreasuryDirect or a bank, broker, or dealer.
- **Cash Management Bills (CMBs)** have variable terms from a few days to one year, are auctioned only when Treasury needs them, and must be bought through a bank, broker, or dealer.

Their scheduling and access differ, but their cash-flow and valuation mechanics are the same.

Sources: Treasury bills; Cash Management Bills.

Treasury Bill Cash Flows



Par Amount, Purchase Price, and Discount

A bill has two investor cash flows: the purchase price $-V_B$ today and the par payment $+V_P$ at maturity.

- **Par amount V_P:** the fixed amount Treasury promises to pay at maturity.
- **Purchase price V_B:** the cash paid to acquire the bill; it is normally below par at auction.
- **Dollar discount $V_P - V_B$:** the interest earned when a below-par bill is held to maturity.

Par determines the maturity payment; price determines the amount invested. The dollar discount is not a rate until a horizon and quotation convention are specified.

Two Ways to Bid at a T-Bill Auction

Treasury uses a single-price auction: every bidder who receives bills pays the same auction price.

- **Noncompetitive bid:** Most individual investors request a par amount and accept the auction-determined rate; within the auction limit, the award is guaranteed. TreasuryDirect offers only this bid type.
- **Competitive bid:** Institutional and other experienced investors request a par amount and state the minimum bank discount rate they will accept: control over the minimum return, at the risk of a partial or no award. These bids go through a bank, broker, or dealer.

Source: TreasuryDirect: How Auctions Work.

A Competitive Bid Controls Allocation, Not Price

Let d^* be the auction cutoff rate: the highest competitive bank discount rate to receive an award. A bidder requesting par amount V_P gets:

$$\begin{aligned} d < d^* &\implies 100\% \text{ of requested par,} \\ d = d^* &\implies \text{possibly a prorated share of requested par,} \\ d > d^* &\implies \text{no award.} \end{aligned}$$

If a bidder requests $10{,}000$ USD of par, a full award is $10{,}000$ USD and a 60% proration is $6{,}000$ USD: award amounts, not cash prices.

Treasury computes one price from d^*, and every awarded bidder pays that price per 100 USD of par.

Source: TreasuryDirect auction allocation and proration.

From Bank Discount Rate to Auction Price

If a bill has D days to maturity, its bank discount rate d implies:

$$\boxed{V_B = V_P \left(1 - d\frac{D}{360}\right)}.$$

Treasury reports the price per 100 USD of par, so set $V_P = 100$ USD.

- A price of 99.72 USD means the investor pays 99.72 USD today for 100 USD at maturity.
- If held to maturity, the dollar interest is $100 - 99.72 = 0.28$ USD per 100 USD of par.

The bank discount convention uses par value and a 360-day year.

Source: TreasuryDirect: Understanding Pricing.

Valuing a T-Bill as an Abstract Asset

Let y be a nominal annual valuation yield, n its compounding frequency, and $g_y = n\log(1 + y/n)$. Setting NPV to zero gives:

$$V_B = \mathcal{D}^{-1}_{T,0}(y;n)\,V_P = \left(1 + \frac{y}{n}\right)^{-nT} V_P = e^{-g_y T} V_P.$$

The price-implied annualized log yield and holding-period log return are:

$$g_B = \frac{1}{T}\log\left(\frac{V_P}{V_B}\right), \qquad r_B = T\,g_B.$$

For the model price, $g_B = g_y$; for an observed market price, g_B is implied by that price and may differ from the model input.

Zero NPV does not mean zero return or zero risk. It means the price reflects the return required by the market under the stated model.

A \$100 T-Bill: Price, Gain, and Return

A 100 USD bill matures in six months. Use nominal annual yield $y = 5\%$ with semiannual compounding, so $n = 2$ and $T = 0.5$ year.

$$\begin{aligned} V_B &= 100(1.025)^{-1} = 97.56098 \approx 97.56 \text{ USD}, \\ V_P - V_B &\approx 2.44 \text{ USD}, \\ g_B &= 2\log(1.025) \approx 4.9385\%/\text{year}, \\ r_B &= \log(1.025) \approx 2.4693\% \text{ over six months.} \end{aligned}$$

Annualized growth and holding-period return are different quantities.

Treasury bank-discount and investment-yield quotations are not automatically y or g_y; match day-count and compounding conventions before comparing.

Formulas: official Treasury auction formulas.

Treasury Notes and Bonds

Treasury notes and bonds are fixed-rate coupon securities that pay interest every six months and repay par at maturity.

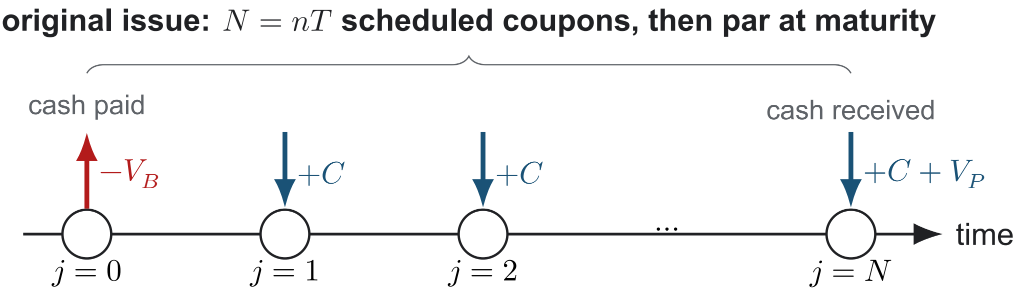
- **Notes:** original terms of 2, 3, 5, 7, or 10 years.
- **Bonds:** original terms of 20 or 30 years.

At original issue, semiannual payments give $n = 2$ and $N = nT$ coupon dates; in a reopening or secondary-market trade, N counts the remaining coupons. With annual coupon rate c, each coupon is $C = (c/n)V_P$:

$$\bar{c}_0 = -V_B, \qquad \bar{c}_j = C \;\; (j = 1, \dots, N-1), \qquad \bar{c}_N = C + V_P.$$

Sources: Treasury notes; Treasury bonds.

Note and Bond Cash Flows



Coupon Rate and Market Yield Do Different Jobs

The coupon rate fixes the promised cash flows; the market yield determines what those cash flows are worth today.

- **Par amount V_P:** fixed principal repaid at maturity.
- **Coupon rate c:** fixed after issue and determines each semiannual payment, $C = (c/2)V_P$.
- **Market yield y:** the return required for the remaining cash flows; it changes with market conditions.
- **Price V_B:** the present value of the remaining coupons and par payment discounted at y, so it changes when y changes.

At an original-issue auction, competitive bidders specify yields, not coupon rates.

The Auction Yield Sets the Coupon and Price

The highest competitive yield to receive an award, the high yield y^*, determines the common auction price: bids below y^* receive the full requested par, bids at y^* may be prorated, and bids above it receive nothing.

After the auction, Treasury chooses a coupon rate in 0.125-percentage-point increments that prices the security closest to, but not above, par.

August 12, 2026 example

- High yield: $y^* = 4.683\%$.
- Coupon rate selected: $c = 4.625\%$.
- Auction price: 99.5407 USD per 100 USD of par.

Sources: TreasuryDirect pricing; 31 CFR 356.20.

Valuing Notes and Bonds as Abstract Assets

Value on a coupon date with N complete coupon periods remaining, and let y be the nominal annual YTM compounded semiannually. Setting NPV to zero and solving for the price gives:

$$0 = \underbrace{-V_B}_{\text{cost}} + \underbrace{\mathcal{D}_{N,0}^{-1}(y;n)V_P}_{\text{par payment}} + \underbrace{C\sum_{j=1}^{N}\mathcal{D}_{j,0}^{-1}(y;n)}_{\text{coupon payments}}$$

$$V_B = \mathcal{D}_{N,0}^{-1}(y;n)V_P + C\sum_{j=1}^{N}\mathcal{D}_{j,0}^{-1}(y;n), \qquad \mathcal{D}_{j,0}^{-1}(y;n) = \left(1+\frac{y}{n}\right)^{-j}.$$

This YTM convention discounts every payment at the same rate. A term-structure valuation instead uses a maturity-specific spot rate for each cash flow.

Are Treasury Securities Risk-Free?

U.S. Treasury securities are commonly treated as low-default-risk nominal-dollar benchmarks. That does not make every government security, or every debt instrument issued in the United States, risk-free.

- **Default risk:** the borrower may miss, suspend, or restructure a promised payment. The central question is: *who promised the payment, and under what terms?*
- **Interest-rate risk:** when market yields change, Treasury prices change. Selling before maturity may realize a gain or loss even when every promised payment is ultimately made.

During financial stress, investors often move toward Treasuries in a flight to quality; the label is relative, not absolute.

Sovereign Debt Can Default or Be Restructured

Government debt is not inherently default-free. Two sovereign examples changed the timing or amount of promised payments:

- **Russia, 1998:** On August 17, Russia announced a de facto default and restructuring of ruble-denominated government bills and bonds; creditor losses amplified an international financial and liquidity crisis.
- **Greece, 2012:** About 97% of the privately held bonds covered by the exchange participated; investors accepted a 53.5% reduction in face value.

Both the probability of default and the recovery after default affect a security's value.

Sources: IMF: Russia; ESM: Greece.

Mortgage Defaults and the 2008 Financial Crisis

The housing collapse shows why the identity of the borrower and the structure of the promised payments matter.

- Falling house prices drove mortgage delinquencies and defaults.
- Losses spread through widely held mortgage-backed securities.
- Leverage and short-term funding amplified the damage and uncertainty about counterparties.
- Investors fled toward U.S. Treasuries in a flight to quality.

Mortgage-backed securities and Treasuries were both dollar-denominated debt but had different borrowers, cash flows, and default risks. The Financial Crisis Inquiry Commission concluded that the crisis was avoidable and warnings were ignored.

Explore: short overview; Financial Crisis Inquiry Commission Report.

Optional Advanced Material

Advanced 1: advanced/default-risk/
CHEME-5660-L2a-Advanced-Bernoulli-DefaultRisk-Fall-2026.ipynb

Optional; not a prerequisite for L2b. The example represents principal and coupon payments with Bernoulli survival indicators, adds recovery after default, and verifies the analytical expected-value price with a Monte Carlo simulation.

Notebooks: lectures/week-2/L2a/advanced, also linked from the lecture notebook.

Summary

This lecture introduced the cash-flow structures, valuation, and risks of U.S. Treasury securities.

Key takeaways

- **Cash flows determine value.** A bill's par amount is its maturity payment, not necessarily its purchase price. Notes and bonds pay coupons and return par. Price puts every cash flow at one valuation date.
- **Coupon rates and yields do different jobs.** The coupon rate fixes promised payments; the market yield discounts them and determines price. Quotation, day-count, compounding, and accrued-interest conventions must match before prices or rates are compared.
- **Risk-free is a relative term.** Treasuries have low nominal default risk and often attract investors during stress, but their prices can still fall when yields rise.

Treasury securities connect dated cash flows, discounting, and market risk.

That’s a wrap. What are some of the interesting things we discussed today?